

174)6

A(2,1,0) B(1,3,-2)

5. Puntak
Zunuk
etc. plawak

Avikitu $\overline{AP} = \frac{1}{3} \overline{AB}$ eta $\overline{AQ} = \frac{2}{3} \overline{AB}$ beteko
daben P eta Q puntuak.

$$\overline{AB} = \overline{OB} - \overline{OA} = (1, 3, -2) - (2, 1, 0) = (-1, 2, -2)$$

$$\overline{AP} = \frac{1}{3} \overline{AB} = \frac{1}{3} (-1, 2, -2)$$

$$\overline{AQ} = \frac{2}{3} \overline{AB} = \frac{2}{3} (-1, 2, -2)$$

$$\begin{aligned} \overline{AQ} &= \overline{OQ} - \overline{OA} \rightarrow \overline{OQ} = \overline{AQ} + \overline{OA} = \\ &= \frac{2}{3} (-1, 2, -2) + (2, 1, 0) \\ &= \left(-\frac{2}{3} + 2, \frac{4}{3} + 1, -\frac{4}{3} + 0 \right) \end{aligned}$$

$$\boxed{\overline{OQ} = \left(\frac{4}{3}, \frac{7}{3}, -\frac{4}{3} \right)}$$

$$\overline{AP} = \overline{OP} - \overline{OA} = \frac{1}{3} (-1, 2, -2)$$

$$\overline{OP} = \overline{AP} + \overline{OA} = \frac{1}{3} (-1, 2, -2) + (2, 1, 0)$$

$$\boxed{\overline{OP} = \left(\frac{5}{3}, \frac{5}{3}, -\frac{2}{3} \right)}$$

174)5

a-reu eta b-reu balizak A(1,2,-1), B(3,0,-2)
eta C(4,a,b) lorokatu egoteko.

Lorokatu egoteko $\overline{AB}$ eta $\overline{BC}$ proportioan
izan behar dira

$$\overline{AB} = \overline{OB} - \overline{OA} = (3, 0, -2) - (1, 2, -1) = (2, -2, -1)$$

$$\overline{BC} = \overline{OC} - \overline{OB} = (4, a, b) - (3, 0, -2) = (1, a, b+2)$$

$$\frac{2}{1} = \frac{-2}{a} = \frac{-1}{b+2}$$

$$\rightarrow \boxed{\begin{matrix} a = -1 \\ b = -5/2 \end{matrix}}$$

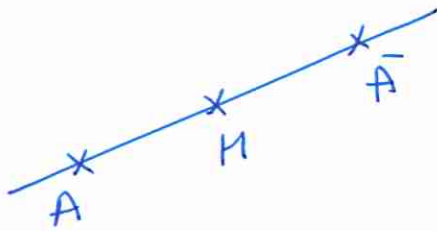
$$\begin{aligned} -2(b+2) &= -a \\ -2b-4 &= 1 \end{aligned}$$

174)2

A-reu simetrikoa π -rekiko

$$A(-2, 3, 0)$$

$$H(1, -1, 2)$$



$$\overrightarrow{AH} = \overrightarrow{HA'}$$

$$\overrightarrow{AA'} = 2\overrightarrow{AH}$$

$$\overrightarrow{AH} = \overrightarrow{OH} - \overrightarrow{OA} = (1, -1, 2) - (-2, 3, 0)$$

$$= (3, -4, 2)$$

$$2\overrightarrow{AH} = 2(3, -4, 2) = (6, -8, 4)$$

$$\overrightarrow{AA'} = 2\overrightarrow{AH} = (6, -8, 4)$$

$$\overrightarrow{AA'} = \overrightarrow{OA'} - \overrightarrow{OA} = (6, -8, 4)$$

$$\overrightarrow{OA'} = (6, -8, 4) + \overrightarrow{OA} = (6, -8, 4) + (-2, 3, 0)$$

$$\boxed{\overrightarrow{OA'} = (4, -5, 4)}$$

Beste modu batena :

$$H = \left(\frac{x_A + x_{A'}}{2}, \frac{y_A + y_{A'}}{2}, \frac{z_A + z_{A'}}{2} \right)$$

$$(1, -1, 2) = \left(\frac{-2 + x_{A'}}{2}, \frac{3 + y_{A'}}{2}, \frac{0 + z_{A'}}{2} \right)$$

$$\left\{ \begin{array}{l} 1 = \frac{-2 + x_{A'}}{2} \longrightarrow x_{A'} = 4 \\ -1 = \frac{3 + y_{A'}}{2} \longrightarrow y_{A'} = -5 \\ 2 = \frac{0 + z_{A'}}{2} \longrightarrow z_{A'} = 4 \end{array} \right\} \boxed{H(4, -5, 4)}$$

(1)

ZUZENAK**(174) 11** koordenatu ardatzen parametrikak eta implizituak.OX ardatzak

$$\vec{a}_r(1,0,0)$$

$$P(0,0,0)$$

PARAMET.

$$\begin{cases} x=1 \\ y=0 \\ z=0 \end{cases}$$

INPLIZ.

$$\begin{cases} y=0 \\ z=0 \end{cases}$$

OY ardatzak

$$\vec{a}_r(0,1,0)$$

$$P(0,0,0)$$

$$\begin{cases} x=0 \\ y=1 \\ z=0 \end{cases}$$

$$\begin{cases} x=0 \\ z=0 \end{cases}$$

OZ ardatzak

$$\vec{a}_r(0,0,1)$$

$$P(0,0,0)$$

$$\begin{cases} x=0 \\ y=0 \\ z=1 \end{cases}$$

$$\begin{cases} x=0 \\ y=0 \end{cases}$$

174)

$$A(-4, 2, 5)$$

15OZ ardatzarekiko $\vec{a}_r(0,0,1)$

$$(x, y, z) = (-4, 2, 5) + \lambda(0, 0, 1)$$

BEKTORIALA

$$\begin{cases} x=-4 \\ y=2 \\ z=5+\lambda \end{cases} \quad \text{PARAMETRIKAK.}$$

$$x = -4 + \lambda \cdot 0 \rightarrow \lambda = \frac{x+4}{0}$$

$$y = 2 + \lambda \cdot 0 \rightarrow \lambda = \frac{y-2}{0}$$

$$z = 5 + \lambda \rightarrow \lambda = \frac{z-5}{1}$$

EK JARRAITZA

$$\frac{x+4}{0} = \frac{y-2}{0} = \frac{z-5}{1}$$

Implizituak.

$$\begin{cases} x+4=0 \\ y-2=0 \end{cases}$$

174)13

$P(1, -3, 0)$
 $\vec{u} \wedge \vec{v}$ vektorekiko paralelo.

$$\vec{u}(1, -1, 2) \quad \vec{v}(2, 0, 0)$$

$$\vec{u} \wedge \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 2 & 0 & 0 \end{vmatrix} = 4\hat{j} + 2\hat{k} \rightarrow (0, 4, 2)$$

$$r: \begin{cases} P(1, -3, 0) \\ \vec{dr}(0, 4, 2) \end{cases}$$

$$(x, y, z) = (1, -3, 0) + \lambda(0, 4, 2) \quad \text{BKT.}$$

$$\text{PARA TI} \rightarrow \begin{cases} x = 1 \\ y = -3 + 4\lambda \\ z = 2\lambda \end{cases}$$

$$\frac{x-1}{0} = \frac{y+3}{4} = \frac{z-0}{2} \quad \text{JARRAIT}$$

$$\begin{aligned} (x-1) \cdot 4 &= 0 \\ (x-1) \cdot 2 &= 0 \\ (y+3) \cdot 2 &= 4(z-0) \\ \Rightarrow y+3 &= 2z \end{aligned} \rightarrow \begin{cases} x-1=0 \\ y-2z=-3 \end{cases} \quad \text{INPLUZATZA}$$

174)14

$$r: \begin{cases} 2x - y + z - 3 = 0 \\ x - y + z - 2 = 0 \end{cases}$$

kalkulatu parametroak

Sistema ebazten dogu:

$$\begin{cases} 2x - y + z = 3 \\ x - y + z = 2 \end{cases}$$

$$\begin{cases} 2x - y = 3 - \lambda \\ x - y = 2 - \lambda \\ z = \lambda \end{cases}$$

$$r: \begin{cases} x = 1 \\ y = \lambda - 1 \\ z = \lambda \end{cases}$$

$$s: \begin{cases} x + 2y - z - 3 = 0 \\ 3y + 2z - 1 = 0 \end{cases}$$

$$\begin{pmatrix} 2 & -1 & 1 & 3-\lambda \\ 1 & -1 & 1 & 2-\lambda \end{pmatrix}$$

$$x = \frac{\begin{vmatrix} 3-\lambda & -1 \\ 2-\lambda & -1 \end{vmatrix}}{-1} = \frac{\cancel{3}-3+2-\cancel{\lambda}}{-1} = 1$$

$$y = \frac{\begin{vmatrix} 2 & 3-\lambda \\ 1 & 2-\lambda \end{vmatrix}}{-1} = \frac{4-2\lambda-3+\lambda}{-1} = \frac{1-\lambda}{-1} = \lambda-1$$

$$z = \lambda$$

$$S: \begin{cases} x + 2y - z = 3 \\ 3y + 2z = 1 \end{cases}$$

$$x = \frac{\begin{vmatrix} 3-1 & 2 \\ 1-2 & 3 \end{vmatrix}}{3} = \frac{9-3-2+4}{3} = \frac{3}{3} = 1$$

$$\begin{cases} x + 2y = 3 - z \\ 3y = 1 - 2z \\ z = 1 \end{cases}$$

$$y = \frac{\begin{vmatrix} 1 & 3-1 \\ 0 & 1-2 \end{vmatrix}}{3} = \frac{1-2}{3} = -\frac{1}{3}$$

$$\begin{vmatrix} 1 & 2 & 1 & 3-1 \\ 0 & 3 & 1 & 1-2 \end{vmatrix}$$

$$r: \begin{cases} x = \frac{4+1}{3} \\ y = \frac{1-2}{3} \\ z = 1 \end{cases}$$

IHAZKO 14

$$a) \begin{cases} x - y = 0 \\ y + z = 2 \end{cases} \rightarrow \begin{matrix} \vec{n}_1(1, -1, 0) \\ \vec{n}_2(0, 1, 1) \end{matrix}$$

$$\vec{d} = \vec{n}_1 \wedge \vec{n}_2 = (1, -1, 0) \wedge (0, 1, 1) =$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = -\hat{i} + \hat{k} - \hat{j}$$

$$\boxed{\vec{d} = (-1, -1, 1)}$$

b) Parametrikoa

$$r: \begin{cases} P \\ \vec{d} \end{cases}$$

zuzenaren puntu bat lotzen dugu.

Adibidez

$$x=0 \rightarrow y=0 \rightarrow z=2 \quad \underline{\underline{P(0,0,2)}}$$

$$\begin{cases} x = 0 - \lambda \\ y = 0 - \lambda \\ z = 2 + \lambda \end{cases}$$

Planuaren
bektore normala
 $\vec{n}(a,b,c)$

174)15

$$r: \frac{x}{2} = \frac{y+1}{-1} = z$$

$$\frac{x}{2} = z \rightarrow x = 2z$$

$$\frac{x}{2} = \frac{y+1}{-1} \rightarrow 2y+2+x=0$$

$$\begin{cases} x - 2z = 0 \\ x + 2y + 2 = 0 \end{cases}$$

ZU ZEHENEN POSITION ERLÄUTEREN

174)17

$$r: \frac{x-1}{3} = \frac{y+2}{2} = \frac{z-1}{4}$$

$$P_r(1, -2, 1) \\ \vec{dr}(3, 2, 4)$$

a)

$$s: \frac{x+2}{-1} = \frac{y-3}{2} = \frac{z-2}{3}$$

$$P_s(-2, 3, 2) \\ \vec{ds}(-1, 2, 3)$$

• Norabide 11? $\frac{3}{-1} \neq \frac{2}{2} \neq \frac{4}{3} \rightarrow$ NORABIDE DES BERDINA

• Gurutzatu edo eba ki?

$$\vec{rP_s} = (-2, 3, 2) - (1, -2, 1) = (-3, 5, 1)$$

$$\begin{vmatrix} \vec{rP_s} \\ \vec{dr} \\ \vec{ds} \end{vmatrix} = \begin{vmatrix} -3 & 5 & 1 \\ 3 & 2 & 4 \\ -1 & 2 & 3 \end{vmatrix} = -51 \neq 0$$

Hiru bektoreak eta diru planukoak dira
beraz plano desberdina da
GURUTZATUEN diru

(3)

$$b) \quad r: \frac{x-1}{-1} = \frac{y-1}{2} = \frac{z-2}{1} \quad s: \frac{x-4}{4} = \frac{y-4}{1} = \frac{z-5}{2}$$

$$P_r(1, 1, 2) \\ \vec{d}_r(-1, 2, 1)$$

$$P_s(4, 4, 5) \\ \vec{d}_s(4, 1, 2)$$

• $\vec{d}_r$ eto $\vec{d}_s$ reu norabideak:

$$-\frac{1}{4} \neq \frac{2}{1} \neq \frac{1}{2} \Rightarrow \text{Norabide desberdino} \begin{cases} \text{Guzutatu} \\ \text{edo} \\ \text{Ebaki} \end{cases}$$

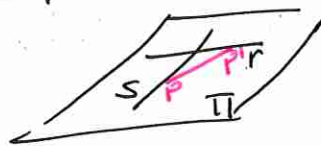
• $\vec{P}_r \vec{P}_s$ bektorea:

$$\vec{P}_r \vec{P}_s = (4, 4, 5) - (1, 1, 2) = (3, 3, 3)$$

Plaukideok diru $\vec{d}_r, \vec{d}_s$ eto $\vec{P}_r \vec{P}_s$?!

$$\begin{vmatrix} -1 & 2 & 1 \\ 4 & 1 & 2 \\ 3 & 3 & 3 \end{vmatrix} = 0 \rightarrow \text{Plaukideok diru}$$

ZUZENAK EBAKITUEN
DIRA



$$c) r: \frac{x}{2} = y-1 = \frac{z+1}{3}$$

$$P_r(0, 1, -1) \\ \vec{d}_r(2, 1, 3)$$

$$s: \begin{cases} x-2y-1=0 \\ 3y-z+1=0 \end{cases} \quad \begin{matrix} x=0 \\ y=-1/2 \\ -3/2 - z + 1 = 0 \\ z = -1/2 \end{matrix}$$

$$P_s(0, -\frac{1}{2}, -\frac{1}{2})$$

$$\vec{d}_s = \vec{n}_1 \wedge \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 0 \\ 0 & 3 & -1 \end{vmatrix}$$

$$2\hat{i} + 3\hat{j} + \hat{k} \\ \vec{d}_s = (2, 3, 1)$$

• $\vec{d}_r$ eto $\vec{d}_s$ reu norabideak!

$$\frac{2}{2} = \frac{1}{1} = \frac{3}{3}$$

Bardinok
diraue?

koitidentetuek
edo
paralelek.

• $\vec{P}_r \in r$ daua aztertuta ea s-u dojoan

$$P_r(0, 1, -1) \rightarrow s: \begin{cases} 0-2 \cdot 1-1 \neq 0 \\ 3 \cdot 1 - (-1) + 1 \neq 0 \end{cases} \text{ Et } \underline{\text{do betituen}}$$

ZUZENAK PARALELOAK diru

$$d) \quad r: \frac{x-1}{2} = \frac{y}{3} = \frac{z}{4}$$

$$P_r(1, 0, 0)$$

$$\vec{dr}(2, 3, 4)$$

$$s: \begin{cases} x = 3 + 4t \\ y = 3 + 6t \\ z = 4 + 8t \end{cases}$$

$$P_s(3, 3, 4)$$

$$\vec{ds}(4, 6, 8)$$

Normalizado atartuta:

~~$$\frac{2}{4} = \frac{3}{6} = \frac{4}{8}$$~~

$$\frac{2}{4} = \frac{3}{6} = \frac{4}{8}$$

direto $\vec{ds}$
proportionalak
dira $\rightarrow$

Normalak berdinak

Zuzenak // edo koinkidentzek

dira bera atartuta dozu ea $P_s \in r$

$$P_s(3, 3, 4) \rightarrow r \left\{ \frac{3-1}{2} = \frac{3}{2} = \frac{4}{4} \checkmark \right.$$

$P_s \in r \rightarrow$ ZUZENA KOINKIDENTZAK DIRA

$$\frac{2(x-1/2)}{3} = \frac{x-1/2}{3/2}$$

174)18

$$r: x=y=z-a$$

$$s: \frac{2x-1}{3} = \frac{y+3}{-2} = \frac{z-2}{0}$$

Ebakitzeko $\vec{dr} \times \vec{ds}$ eta $\vec{dr}, \vec{ds}$ eta $\vec{P_r P_s}$
planokideak izan behar dira, bera euren
determinantea = 0.

$$\vec{dr} = (1, 1, 1)$$

$$P_r = (0, 0, a)$$

$$\vec{ds} = (3/2, -2, 0)$$

$$P_s = (1/2, -3, 2)$$

$$\vec{P_r P_s} = \left(\frac{1}{2}, -3, 2 \right) - (0, 0, a) = \left(\frac{1}{2}, -3, 2-a \right)$$

$$\begin{vmatrix} \frac{1}{2} & -3 & 2-a \\ 1 & 1 & 1 \\ 3/2 & -2 & 0 \end{vmatrix} = \frac{-9}{2} - 2(2-a) - \frac{3}{2}(2-a) + 1 =$$

$$= \frac{-9}{2} - 4 + 2a - 3 + \frac{3a}{2} + 1 = \frac{5a}{2} - \frac{21}{2} = 0$$

$$\boxed{a=3}$$

Ebakite puntus kalkulatzeko $a=3$ dauka: ④

$$r: x=y=z-3$$

$$s: \frac{2x-1}{3} = \frac{y+3}{-2} = \frac{z-2}{0}$$

PARAMETRIZETARA pasatuz:

$$r: \begin{cases} x = \lambda \\ y = \lambda \\ z = \lambda + 3 \end{cases}$$

$$s: \begin{cases} \frac{2x-1}{3}: \mu & x = \frac{1}{2} + \frac{3}{2}\mu \\ \frac{y+3}{-2}: \mu & y = -3 - 2\mu \\ \frac{z-2}{0}: \mu & z = 2 \end{cases}$$

$$\begin{array}{lcl} x \text{ berdinduz} \rightarrow & \lambda = \frac{1}{2} + \frac{3}{2}\mu & \\ y \text{ " } \rightarrow & \lambda = -3 - 2\mu & \\ z \text{ " } \rightarrow & \lambda + 3 = 2 & \end{array} \left\{ \begin{array}{l} \lambda - \frac{3}{2}\mu = \frac{1}{2} \\ \lambda + 2\mu = -3 \\ \lambda = -1 \end{array} \right.$$

$$\lambda = -1 \left\{ \begin{array}{l} \mu = \frac{(-3+1)}{2} = -1 \\ \mu = \left(\frac{1}{2} + 1\right)\left(-\frac{2}{3}\right) = \frac{3}{2}\left(-\frac{2}{3}\right) = -1 \end{array} \right.$$

$$\boxed{\lambda = -1 \rightarrow \mu = -1} \rightarrow r \text{ eta } s \text{ -u.}$$

$$r: \begin{cases} x = -1 \\ y = -1 \\ z = 2 \end{cases}$$

$$s: \begin{cases} x = \frac{1}{2} - \frac{3}{2} = -1 \\ y = -3 - 2(-1) = -1 \\ z = 2 \end{cases}$$

$$\boxed{P(-1, -1, 2)}$$

RI ZUZENEN BERTID ERLANBOA

1.) KOMPARTU ea $\vec{dr}$ eta $\vec{ds}$ NORABIDE BERIDEXKAK DIREN

$$\frac{dr_x}{ds_x} = \frac{dr_y}{ds_y} = \frac{dr_z}{ds_z}$$

NORABIDE BERDINA $\leq$

• Propotionalak badina.

• Aztertuta $P_r \in s$.

$P_r \in s$ —
 $P_r \notin s$ =

NORABIDE DIBERDINTA

Aztertuta ea ebaki
 ala gurutzatu diren.

(hirugarren bektoreak) alaba $\vec{P_r P_s}$

Aztertuta ea planokideak diren $\vec{dr}, \vec{ds}$ $\vec{P_r P_s}$



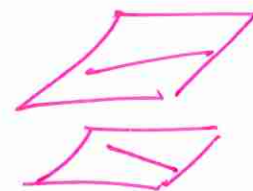
$$\begin{vmatrix} dr_x & dr_y & dr_z \\ ds_x & ds_y & ds_z \\ \vec{P_r P_s}_x & \vec{P_r P_s}_y & \vec{P_r P_s}_z \end{vmatrix}$$

$= 0$ Planokideak



$\neq 0$

Plano
 desberdinak
 GURUTZATU



174)19

Kalkulatu m paralelbox izateko:

(5)

$$r: \begin{cases} x = 5 + 4t \\ y = 3 + t \\ z = -t \end{cases}$$

$$P(5, 3, 0) \\ \vec{dr} (4, 1, -1)$$

$$s: \frac{x}{m} = \frac{y-1}{3} = \frac{z+3}{n}$$

$$P(0, 1, -3) \\ \vec{ds} (m, 3, n)$$

• PARALELOAK BADIRA $\rightarrow \vec{dr}$ eta $\vec{ds}$ posizioak dira.

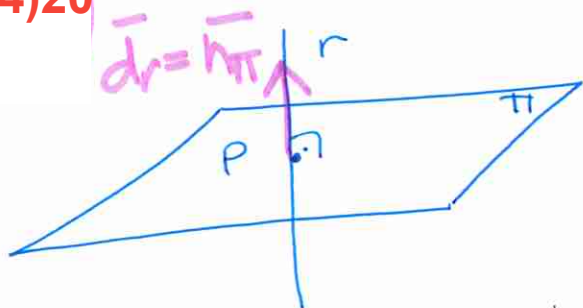
$$\frac{4}{m} = \frac{1}{3} = \frac{-1}{n} \Rightarrow \boxed{\begin{matrix} m=12 \\ n=-3 \end{matrix}}$$

• P_r alderatu da $ea \in S$ dagoen.

$$P(5, 3, 0) \rightarrow \frac{5}{12} \neq \frac{3-1}{3} \neq \frac{0+3}{-3} \quad \text{Berat } \boxed{P_r \notin S}$$

PARALELOAK dira

174)20



$$\pi: x + 2y = 5 \rightarrow \vec{n}_\pi = (1, 2, 0) \\ P(7, -2, 1)$$

Kalkulatuko π planoarekiko zuzen perpendikularra bere puntu eta $\vec{dr}$ behar da (π planoareko normala).

$$P(7, -2, 1) \\ \vec{dr} = \vec{n}_\pi = (1, 2, 0)$$

$$r: \begin{cases} x = 7 + t \\ y = -2 + 2t \\ z = 1 \end{cases} \Rightarrow \frac{x-7}{1} = \frac{y+2}{2} = \frac{z-1}{0}$$

IHAZKOA

$$S: \begin{cases} P(-1, 5, 0) \\ r \text{ zuzuenaren paraleloa} \end{cases} \quad r: \begin{cases} x+y=1 \rightarrow \vec{n}_1=(1, 1, 0) \\ y-z=2 \rightarrow \vec{n}_2=(0, 1, -1) \end{cases}$$

S zuzuenaren norabide bektorea $\vec{ds}$, $\vec{dr}$ -ren bardina da paraleloa direlako.

$$\begin{aligned} \vec{dr} &= \vec{n}_1 \wedge \vec{n}_2 = (1, 1, 0) \wedge (0, 1, -1) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} \\ &= -\hat{i} + \hat{k} + \hat{j} = (-1, 1, 1) \end{aligned}$$

$$\text{Beraz } S \text{ zuzena } \begin{cases} P(-1, 5, 0) \\ \vec{ds} = (-1, 1, 1) \end{cases}$$

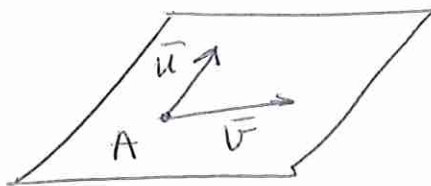
$$S: \begin{cases} x = -1 + \lambda \\ y = 5 + \lambda \\ z = 0 + \lambda \end{cases} \Rightarrow \frac{x+1}{-1} = \frac{y-5}{1} = \frac{z}{1}$$

PLANOAK

175)22

INPLIZITAK

a/ $A(1, -3, 2)$
 $\vec{u}(2, 1, 0)$
 $\vec{v}(-1, 0, 3)$



Planoaren
 ekuazioa
 bi modutara
 kalkulatu da.

1.) MODUA

$$(x, y, z) = (1, -3, 2) + \lambda(2, 1, 0) + \mu(-1, 0, 3) \quad \text{Bektoren}$$

$$\begin{cases} x = 1 + 2\lambda - \mu \\ y = -3 + \lambda \\ z = 2 + 3\mu \end{cases} \quad \text{Parametrikoki}$$

Inplizitu bidez λ eta μ ezabatu:

$$\begin{cases} 2\lambda - \mu = x - 1 \\ \lambda = y + 3 \\ 3\mu = z - 2 \end{cases} \quad \begin{vmatrix} 2 & -1 & x-1 \\ 1 & 0 & y+3 \\ 0 & 3 & z-2 \end{vmatrix} = 0$$

$$3(x-1) + z-2 - 6(y+3) = 0$$

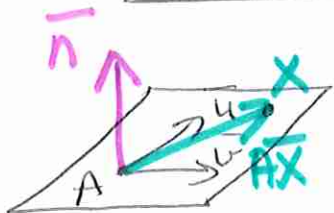
$$3x - 3 + z - 2 - 6y - 18 = 0$$

$$\boxed{\pi: 3x - 6y + z - 23 = 0}$$

2. MODUA

$$\vec{n} = \vec{u} \wedge \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 0 \\ -1 & 0 & 3 \end{vmatrix} = 3\hat{i} + \hat{k} - 6\hat{j}$$

$$\vec{n} = (3, -6, 1)$$



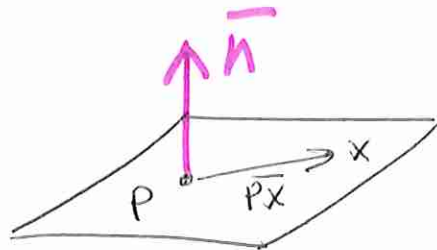
$$\vec{n} \perp \vec{AX} \rightarrow (3, -6, 1) \cdot (x-1, y+3, z-2) = 0$$

$$3(x-1) - 6(y+3) + z-2 = 0$$

$$\vec{AX} = (x-1, y+3, z-2)$$

$$\boxed{\pi: 3x - 6y + z - 23 = 0}$$

$$b) \pi: \begin{cases} P(2, -3, 1) \\ \bar{n}(5, -3, -4) \end{cases}$$



$$\bar{n} \perp \bar{P}x \rightarrow \bar{n} \cdot \bar{P}x = 0.$$

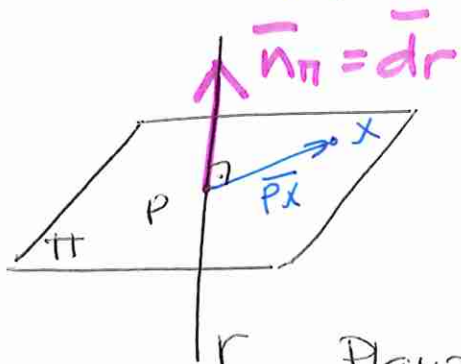
$$\bar{P}x = (x-2, y+3, z-1).$$

$$(5, -3, -4) \cdot (x-2, y+3, z-1) = 0.$$

$$5(x-2) + (-3)(y+3) - 4(z-1) = 0$$

$$\boxed{\pi: 5x - 3y - 4z - 15 = 0}$$

$$c) \begin{cases} \pi \perp r \\ P(1, 0, 1) \end{cases} \quad r: \frac{x}{2} = \frac{y+1}{-1} = \frac{z}{3}.$$



r vektorearen $\bar{d}r$, planoaren
vektore normala da:

$$\bar{d}r = \bar{n}_{\pi}$$

$$\bar{d}r = (2, -1, 3) = \bar{n}_{\pi}.$$

Planoaren ekuazioa kalkulatzeko.

$$\bar{n} \perp \bar{P}x \quad \bar{P}x = (x-1, y, z-1)$$

$$\bar{n} \cdot \bar{P}x = 0$$

$$(2, -1, 3) \cdot (x-1, y, z-1) = 0$$

$$2(x-1) - y + 3(z-1) = 0.$$

$$2x - 2 - y + 3z - 3 = 0.$$

$$\pi: \boxed{2x - y + 3z - 5 = 0}$$

EKUAZIO PARAMETRIKAK eta INPLIZITAK

X4 PLANOA

$$\begin{cases} x=1 \\ y=1 \\ z=0 \end{cases}$$

$$z=0$$

Y2 PLANOA

$$\begin{cases} x=0 \\ y=1 \\ z=1 \end{cases}$$

$$x=0$$

X2 PLANOA

$$\begin{cases} x=1 \\ y=0 \\ z=1 \end{cases}$$

$$y=0$$

175)23 PLANEN EK. PARAMETRIKAK

a) $z=3$

$$\begin{cases} x=1 \\ y=1 \\ z=3 \end{cases}$$

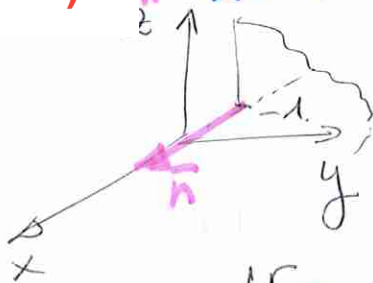
b) $x=-1$

$$\begin{cases} x=-1 \\ y=1 \\ z=1 \end{cases}$$

c) $y=2$

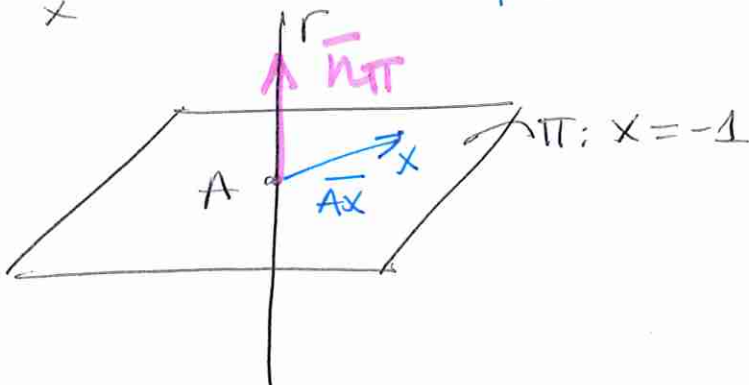
$$\begin{cases} x=1 \\ y=2 \\ z=1 \end{cases}$$

175)24 $x=-1$ planaren bektore normale.



$$\vec{n} = (1, 0, 0)$$

$r \mid x=-1$ planarekin r perpendikulara
 $A(2, 3, 0)$



Planaren bektore normale, zuzena $\vec{r}$ da.

$$r \mid \begin{cases} A(2, 3, 0) \\ \vec{r} = (1, 0, 0) \end{cases}$$

$$r: \begin{cases} x=2+1 \\ y=3 \\ z=0 \end{cases} \Rightarrow$$

$$x-2 = \frac{y-3}{0} = \frac{z}{0}$$

175)25

kalkulatu m eta n, paraleloak izateko.

$$\alpha: mx + y - 3z - 1 = 0 \rightarrow \bar{n}_\alpha = (m, 1, -3)$$

$$\beta: 2x + ny - z - 3 = 0 \rightarrow \bar{n}_\beta = (2, n, -1)$$

Izan daitezke koinkidenteak?

Planok PARALELOAK izateko $\rightarrow$ Bektore normalak proportionalak

$$\frac{m}{2} = \frac{1}{n} = \frac{-3}{-1} \rightarrow \boxed{\begin{matrix} m=6 \\ n=1/3 \end{matrix}}$$

$$\left\{ \begin{array}{l} \alpha: 6x + y - 3z - 1 = 0 \\ \beta: 2x + \frac{1}{3}y - z - 3 = 0 \end{array} \right\} \leftarrow \text{paraleloak}$$

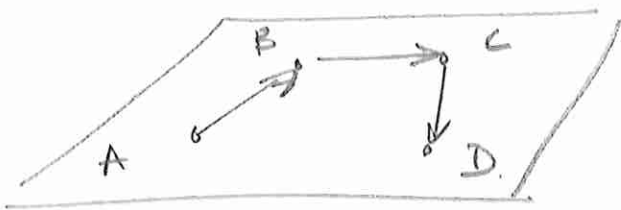
Paraleloak dira eta ez koinkidenteak
 goi gutxiak proportionalak direlako
 goi orkeo izan ean.

$$\frac{6}{2} = \frac{1}{1/3} = \frac{-3}{-1} \quad \left(\neq \frac{-1}{-3} \right) !!$$

175)26

A(0,1,-2) B(1,2,0) C(0,0,1) D(1,0,m)

kalkulatu m plano kided izateko



lau puntuen artean 3 bektore
 sortu, osatutako motzreoren
 heine, 2 izan behar da

$$\overline{AB} = (1, 2, 0) - (0, 1, -2) = (1, 1, 2) \quad \text{Determin} = 0, 2 \text{ bektore}$$

$$\overline{BC} = (0, 0, 1) - (1, 2, 0) = (-1, -2, 1) \quad \text{bakteri LI izateko.}$$

$$\overline{CD} = (1, 0, m) - (0, 0, 1) = (1, 0, m-1)$$

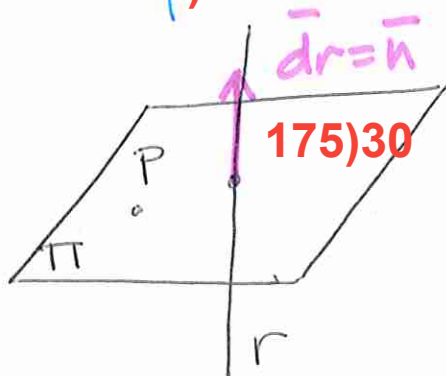
$$\begin{vmatrix} 1 & 1 & 2 \\ -1 & -2 & 1 \\ 1 & 0 & m-1 \end{vmatrix} = -2(m-1) + 1 + 4 + m-1 = -2m + 2 + 1 + 4 + m - 1 = -m + 6 = 0$$

$$\boxed{m=6}$$

175)30)

$$\left. \begin{array}{l} \pi \perp r: \\ \pi \end{array} \right\} P(1, -4, 2)$$

175)30



175)30

$$r: \begin{cases} x - 2y + 6 = 0 \\ 3x + 2y - 4 = 0 \end{cases}$$

π plano kolkulatuکو پونئو
eta $\bar{n}_\pi$ belor doju (zuzenoren
uzrobide bektorea
izaugo dela)

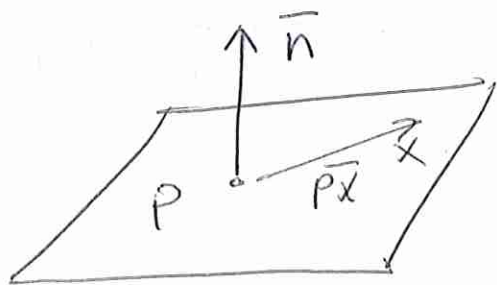
Zuzenoren uzrobide bektorea aterotako implizitetsh...

$$r: \begin{cases} x - 2y + 6 = 0 \rightarrow \bar{n}_1 = (1, -2, 0) \\ 3x + 2y - 4 = 0 \rightarrow \bar{n}_2 = (3, 2, 0) \end{cases}$$

$$\bar{dr} = \bar{n}_1 \wedge \bar{n}_2 = (1, -2, 0) \wedge (3, 2, 0) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 0 \\ 3 & 2 & 0 \end{vmatrix} =$$

$$= 8\hat{k} = (0, 0, 8).$$

Planoa kolkulatuکو π : $\begin{cases} P(1, -4, 2) \\ \bar{n}_\pi = \bar{dr} = (0, 0, 8) \end{cases}$



$$\bar{n} \perp \overrightarrow{PX} \rightarrow \bar{n} \cdot \overrightarrow{PX} = 0.$$

$$\overrightarrow{PX} = (x-1, y+4, z-2)$$

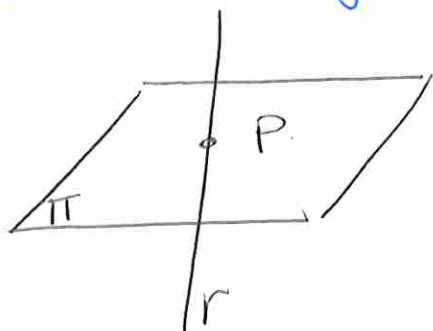
$$(0, 0, 8) \cdot (x-1, y+4, z-2) = 0$$

$$8(z-2) = 0$$

$$\boxed{\pi: z-2=0}$$

175)31

$$r: \frac{x}{2} = y - 3 = \frac{z-2}{-3} \text{ ekuations dau } \pi: x - 2y + z = 1$$



kalkulatu eta ebaki puntu
kalkulatu.

$$\begin{aligned} & r \text{ zuzen parametroetan. } \left\{ \begin{array}{l} x = 2t \\ y = 3 + t \\ z = 2 - 3t \end{array} \right. \quad P(2t, 3+t, 2-3t) \\ & P \in r \\ & \text{Ordezkatzen da puntu} \\ & \text{orokor bat } \pi \text{ planoa.} \rightarrow P \in \pi \text{ probatzeko.} \end{aligned}$$

$$2t - 2(3+t) + 2 - 3t = 1.$$

$$2t - 6 - 2t + 2 - 3t = 1$$

$$-3t - 5 = 0$$

$$t = -5/3. \rightarrow \text{Soluzioa dagoa eta } P \text{ puntuak ebakitzen dau.}$$

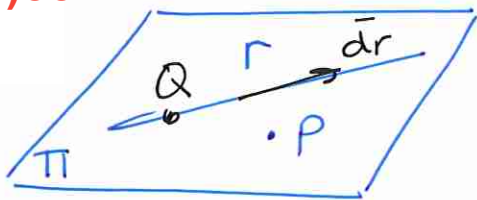
$$x = -10/3$$

$$y = 3 - 5/3 = 4/3$$

$$z = 2 - 3(-5/3) = 2 + 5 = 7.$$

$$\rightarrow \boxed{P(-10/3, 4/3, 7)}$$

175)33A



kalkulatu π parametroetan.

$$P(2, 1, 2)$$

$$r: x - 2 = \frac{y - 3}{-1} = \frac{z - 4}{-3}$$

$$\vec{dr} = (1, -1, -3)$$

$$D = (2, 3, 4)$$

Planorek ekuazio bertales Puntu bat eta bi bektore
behar dira

$$Q(2, 3, 4)$$

$$\vec{PQ} = (2, 3, 4) - (2, 1, 2) = (0, 2, 2)$$

$$\vec{dr} = (1, -1, -3)$$

$$x = 2 + t$$

$$y = 3 - t + 2\mu$$

$$z = 4 - 3t + 2\mu$$

$$r: \frac{x-1}{2} = y = z-2$$

$$\vec{dr} = (2, 1, 1) \quad P(1, 0, 2)$$

175)32

$$s: \begin{cases} x - 2z = 5 \\ x - 2y = 11 \end{cases}$$

$$\vec{ds} = \vec{n}_1 \wedge \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -2 \\ 1 & -2 & 0 \end{vmatrix}$$

$$\vec{ds} = (-4, -2, -2)$$

a) konprobatu || dirau:

Paraleloak dirau jakiteko atxeratu da ea $\vec{ds}$ eto $\vec{dr}$ proportionalak dirau.

$$\vec{dr} = (2, 1, 1)$$

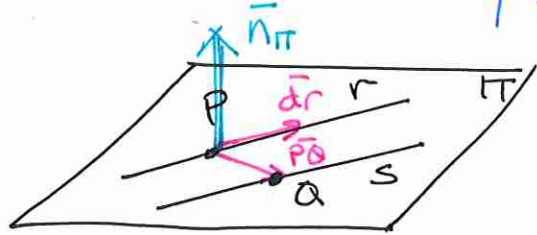
$$\vec{ds} = (-4, -2, -2)$$

$$\frac{2}{-4} = \frac{1}{-2} = \frac{1}{-2} \Rightarrow \text{Proportional. dirau}$$

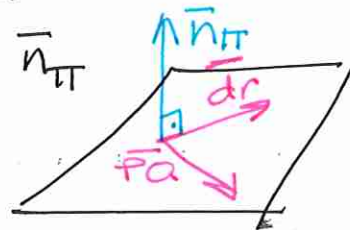
Jakiteko PARALELOAK edo KONTZIDENTEAK dirau atxeratu da ea P puntua ($P \in r$), S azalarean dagoan. $P(1, 0, 2) \rightarrow s: \begin{cases} 1 - 2 \cdot 2 \neq 5 \\ 1 - 2 \cdot 0 \neq 11 \end{cases} \rightarrow \underline{\underline{P \notin S}}$

$P \notin S \rightarrow$ ZUZEN PARALELOAK DIRA

b) π Planoa $\begin{cases} r \in \pi \\ s \in S \end{cases}$



Planorean bi bektore lortu etkeno biekiko $\perp$ dau beste bektore bat lortu ariteke $\vec{n}_\pi$ lortu daiteke.



Lortu da bektore bati bat $\vec{PQ}$

$$Q \in S \rightarrow x=0 \rightarrow z=-5/2 \rightarrow y=-11/2$$

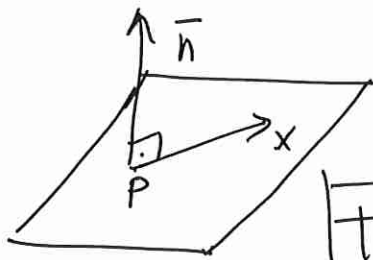
$$\vec{PQ} = (0 - \frac{1}{2}, -\frac{5}{2}) - (1, 0, 2) = (-\frac{1}{2}, -\frac{11}{2}, -\frac{9}{2})$$

$$\vec{n}_\pi = \vec{PQ} \wedge \vec{dr} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -11/2 & -9/2 \\ 2 & 1 & 1 \end{vmatrix} = -\frac{1}{2}\hat{i} - 9\hat{j} - \hat{k} + 11\hat{k} + \hat{j} + \frac{9}{2}\hat{i}$$

$$\vec{n}_\pi = \underline{\underline{(-1, -8, 10)}}$$

$$= -\hat{i} - 8\hat{j} + 10\hat{k}$$

$$\vec{n} \perp \vec{PX} \rightarrow \vec{n} \cdot \vec{PX} = 0$$



$$(-1, -8, 10) \cdot (x-1, y, z-2) = 0$$

$$-x+1-8y+10z-20=0$$

$$\Pi: -x-8y+10z-19=0$$

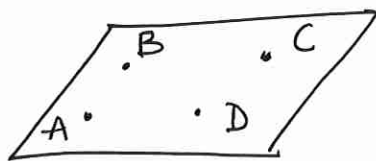
175)31 Planoan dagot A, B, C, eto D?

$$A(1, 0, 0)$$

$$B(0, 1, 0)$$

$$C(2, 1, 0)$$

$$D(-1, 2, 1)$$



Atvratu ai
u pūntuēkiu 3 bektore
eta atvratu da
euvē artiko matītesra
heine.

$$\vec{AD} = (-1, 2, 1) - (1, 0, 0) = (-2, 2, 1)$$

$$\vec{AB} = (0, 1, 0) - (1, 0, 0) = (-1, 1, 0)$$

$$\vec{CD} = (-1, 2, 1) - (2, 1, 0) = (-3, 1, 1)$$

$$\begin{vmatrix} -2 & 2 & 1 \\ -1 & 1 & 0 \\ -3 & 1 & 1 \end{vmatrix} = -2(-1+3) = 2 \neq 0$$

$$\text{rang}(M) = 3 \rightarrow 3 \text{ bektore LI}$$

et dir
plaukideak

175)34

TIRU PLANDEN ARTIKO POSIZIA (10)

$$a) \begin{cases} x + 2y - z - 3 = 0 \\ 3y + 2z - 1 = 0 \\ x + y + z - 2 = 0 \end{cases}$$

$$M' = \begin{pmatrix} 1 & 2 & -1 & | & 3 \\ 0 & 3 & 2 & | & 1 \\ 1 & 1 & 1 & | & 2 \end{pmatrix}$$

n

$$|M| = 8 \quad \text{heiu}(n) = \text{heiu}(n') = 3$$

tiru planoeek puntu batuan ebakitzen dira.

$$b) \begin{cases} 2x - y + z = 3 \\ x - y + z = 2 \\ 3x - y + z = 4 \end{cases}$$

$$M' = \begin{pmatrix} 2 & -1 & 1 & | & 3 \\ 1 & -1 & 1 & | & 2 \\ 3 & -1 & 1 & | & 4 \end{pmatrix}$$

$$n \rightarrow \begin{vmatrix} 2 & -1 \\ 1 & -1 \end{vmatrix} = -2 + 1 = -1$$

$$\begin{vmatrix} 2 & -1 & 1 \\ 1 & -1 & 1 \\ 3 & -1 & 1 \end{vmatrix} = -2 - 3 - 1 + 3 + 1 + 2 = 0.$$

$$n' \rightarrow \begin{vmatrix} 2 & -1 & 3 \\ 1 & -1 & 2 \\ 3 & -1 & 4 \end{vmatrix} = -8 - 3 + 9 + 4 + 4 = 0.$$

$\left\{ \begin{array}{l} \text{heiu}(n) = \text{heiu}(n') \\ = 2 \\ \text{Bat indatun.} \\ \text{PLANOAK} \\ \text{EUKAN} \\ \text{EBAKITZEN} \\ \text{DIRA} \\ \text{ZUZEN BATEN} \end{array} \right.$

$$c) \begin{cases} x - y + z - 1 = 0 \\ 3x + y - 2z = 0 \\ 2x + 2y - 3z + 4 = 0 \end{cases}$$

$$M' = \begin{pmatrix} 1 & -1 & 1 & | & 1 \\ 3 & 1 & -2 & | & 0 \\ 2 & 2 & -3 & | & -4 \end{pmatrix}$$

n

$$n \rightarrow \begin{vmatrix} 1 & -1 \\ 3 & 1 \end{vmatrix} = 1 + 3 = 4$$

$$\begin{vmatrix} 1 & -1 & 1 \\ 3 & 1 & -2 \\ 2 & 2 & -3 \end{vmatrix} = -3 + 4 + 6 - 2 - 9 + 4 = 0$$

$$n' \rightarrow \begin{vmatrix} 1 & -1 & 1 \\ 3 & 1 & 0 \\ 2 & 2 & -4 \end{vmatrix} = -12 \neq 0$$

$\left\{ \begin{array}{l} \text{Batorcelio} \\ \uparrow \\ \text{heiu}(n) = 2 \neq \text{heiu}(n') = 3 \\ \text{PLANOAK BINAN} \\ \text{BINAN EBAKITZEN DIRA} \end{array} \right.$

$$d) \begin{cases} x+y-2z = 0 \\ -x+y+z = -1 \\ 2x-z = -2 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ -1 & 1 & 1 & -1 \\ 2 & 0 & -1 & -2 \end{array} \right) \quad \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 2$$

$$\begin{vmatrix} 1 & 1 & -2 \\ -1 & 1 & 1 \\ 2 & 0 & -1 \end{vmatrix} = -1 + 2 + 4 - 1 \neq 0$$

$$\begin{vmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 2 & 0 & -2 \end{vmatrix} = -2 - 2 - 2 + 1 = -5$$

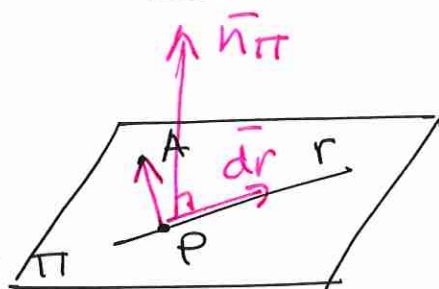
kein $\pi = 3$. = kein π'

Planoak puntu batzua ebakitzen dira

175)33B

A(1,0,1)

$$r: \begin{cases} x+y-z+1=0 \\ 2x-y+2z=0 \end{cases}$$



- Planooren ek. $\bar{u}_\pi$ eto Puntu bat behar da.
- $\bar{u}_\pi$ kalkulatzeko planoaren bi bektore behar dira. $\bar{dr}$ eta $\bar{PA}$, non P r-ren puntu bat dau.
- $\bar{dr}$ ateratzeko r-ren inplizitatuak:

$$\begin{aligned} \bar{dr} &= \bar{u}_1 \wedge \bar{u}_2 = (1, 1, -1) \wedge (2, -1, 2) = \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 2 & -1 & 2 \end{vmatrix} = 2\hat{i} - 2\hat{j} - \hat{k} - 2\hat{k} - 2\hat{j} - \hat{i} = \\ &= \hat{i} - 4\hat{j} - 3\hat{k} = \underline{\underline{(1, -4, -3)}} = \bar{dr} \end{aligned}$$

- P puntua:

$$x=0$$

$$y-z+1=0$$

$$-y+2z=0$$

$$z+1=0$$

$$z=-1$$

$$y=-2$$

$$P(0, -2, -1)$$

$$\bar{PA} = (1, 0, 1) - (0, -2, -1) = \underline{\underline{(1, 2, 2)}} = \bar{PA}$$

• $\bar{n}_\pi$ kalkulatu:

$$\bar{n}_\pi = \overrightarrow{PA} \wedge \overrightarrow{dr} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 1 & -4 & -3 \end{vmatrix} =$$

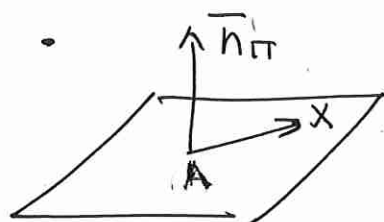
$$= -6\hat{i} + 2\hat{j} - 4\hat{k} - 2\hat{k} + 3\hat{j} + 8\hat{i} =$$

$$= 2\hat{i} + 5\hat{j} - 6\hat{k}$$

$$\bar{n}_\pi = (2, 5, -6)$$

Produkta misto:

$$\begin{array}{l} \overrightarrow{AX} \\ \overrightarrow{PA} \\ \overrightarrow{dr} \end{array} \begin{vmatrix} x-1 & y & z-1 \\ 1 & 2 & 2 \\ 1 & -4 & -3 \end{vmatrix}$$



$$A(1, 0, 1)$$

$$\bar{n} \perp \overrightarrow{AX} \rightarrow \bar{n} \cdot \overrightarrow{AX} = 0.$$

$$(2, 5, -6) \cdot (x-1, y, z-1) = 0$$

$$2(x-1) + 5y - 6(z-1) = 0.$$

$$2x - 2 + 5y - 6z + 6 = 0$$

$$\boxed{\pi: 2x + 5y - 6z + 4 = 0.}$$

175)35

r eta π -ren posizio erlatiboa:

$$r: \begin{cases} x = -t \\ y = 0 \\ z = 1+t \end{cases} \quad t \in \mathbb{R}$$

$$\overrightarrow{dr} = (-1, 0, 1)$$

$$\pi: 2x - y + 3z = 6.$$

$$\bar{n}_\pi = (2, -1, 3)$$

$$\overrightarrow{dr} \cdot \bar{n}_\pi = (-1, 0, 1) \cdot (2, -1, 3) = -2 + 0 + 3 = 1 \neq 0.$$

$\overrightarrow{dr} \cdot \bar{n}_\pi \neq 0$ dauzat r eta π ez dira paraleloak.
beraz EBAKITZEN dira.

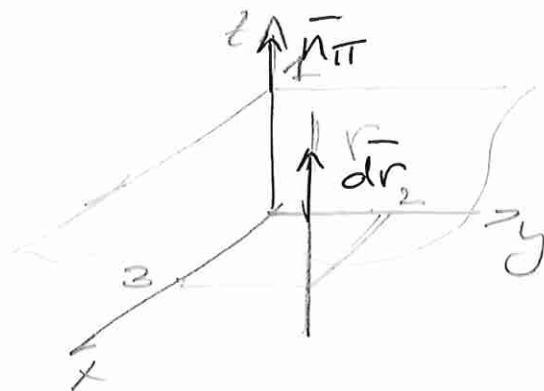
Posizio ERLATIBOA.

$$r: \begin{cases} x=3 \\ y=2 \end{cases}$$

$$\pi: z=1$$

$$\bar{n}_\pi = (0, 0, 1)$$

$$\bar{dr} = (0, 0, 1)$$



Zuzena eta planoa perpendikularrak dira eta ebakitze puntua $(3, 2, 1)$ da.

177)61

$$r: \begin{cases} 3x - 2y + 1 = 0 \\ 2y + 3z - 3 = 0 \end{cases}$$

$$A(1, -3, 2)$$

$$B(0, 1, 1)$$

Planoa aterotzeko puntua eta $\bar{n}_\pi$ behar da

$$\bar{dr} = (3, -2, 0) \wedge (0, 2, 3) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 0 \\ 0 & 2 & 3 \end{vmatrix} =$$

$$= -6\hat{i} + 6\hat{k} - 9\hat{j} = (-6, -9, 6) // (2, 3, -2)$$

$$\bar{u}_\pi = \bar{dr} \wedge \bar{AB}$$

$$\bar{AB} = (0, 1, 1) - (1, -3, 2) = (-1, 4, -1)$$

Bidaktore murrizte

$$\begin{array}{l} \bar{AX} \\ \bar{dr} \\ \bar{AB} \end{array} \begin{vmatrix} x-1 & y+3 & z-2 \\ 2 & 3 & -2 \\ -1 & 4 & -1 \end{vmatrix}$$

$$\begin{aligned} \rightarrow \bar{u}_\pi = \bar{dr} \wedge \bar{AB} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -2 \\ -1 & 4 & -1 \end{vmatrix} = -3\hat{i} + 2\hat{j} + 8\hat{k} \\ &\quad + 3\hat{k} + 2\hat{j} + 8\hat{i} \\ &= 5\hat{i} + 4\hat{j} + 11\hat{k} \\ &= \underline{\underline{(5, 4, 11)}} \end{aligned}$$

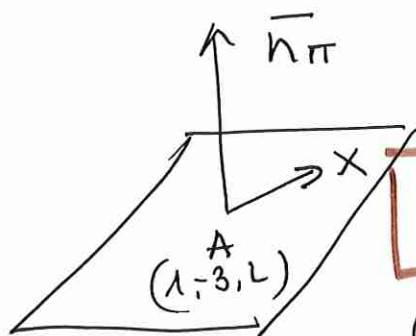
Planoa aterotzeko:

$$\bar{n} \cdot \bar{AX} = 0$$

$$(5, 4, 11) \cdot (x-1, y+3, z-2) = 0$$

$$5x - 5 + 4y + 12 + 11z - 22 = 0$$

$$\pi: \underline{\underline{5x + 4y + 11z - 15 = 0}}$$



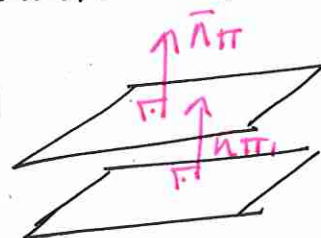
175)29 kalkulasi m plawak paralel atə perpendicular izatiko. $-4m-4=0$
 $n=$

$$\pi: mx + 2y - 3z - 1 = 0 \quad \bar{n}_\pi = (m, 2, -3)$$

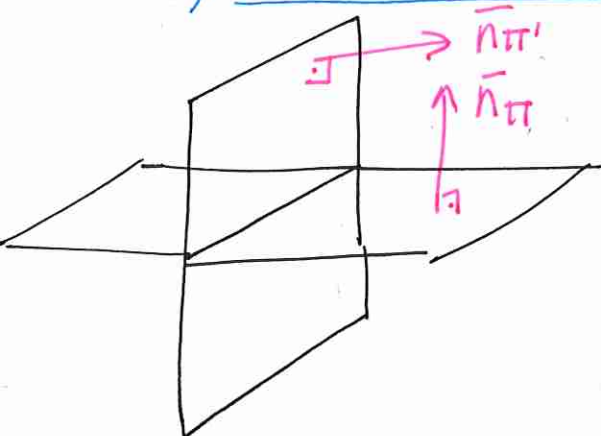
$$\pi': 2x - 4y + 6z + 5 = 0 \quad \bar{n}_{\pi'} = (2, -4, 6)$$

a) PARALELOAK

Paralel izatiko $\bar{n}_\pi = \alpha \bar{n}_{\pi'}$ proporsionalak izan behar dira. $\frac{m}{2} = \frac{2}{-4} = \frac{-3}{6} \rightarrow \boxed{m = -1}$



b) PERPENDIKULARAK



Plawak perpendicularak badira $\bar{n}_\pi \perp \bar{n}_{\pi'}$, beraz

$$\bar{n}_\pi \cdot \bar{n}_{\pi'} = 0.$$

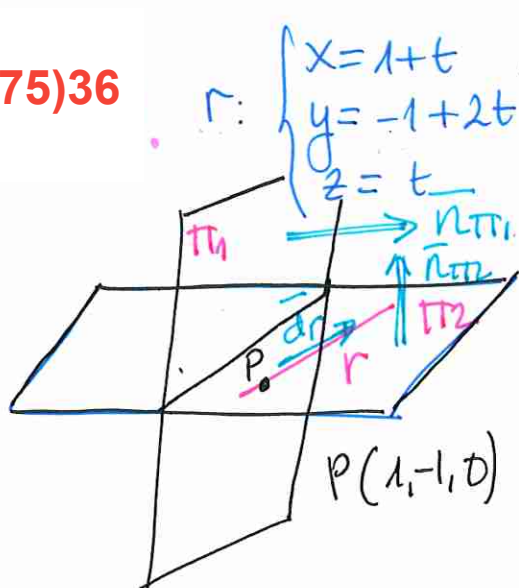
$$(m, 2, -3) \cdot (2, -4, 6) = 0.$$

$$2m - 8 - 18 = 0$$

$$2m - 26 = 0$$

$$\boxed{m = 13}$$

175)36



$$\bar{dr} = (1, 2, 1)$$

$$\pi_1: 2x + y - z - 2 = 0 \quad \bar{n}_1 = (2, 1, -1)$$

kalkulasi π_2

$$\pi_2 \perp \pi_1.$$

$$\bar{n}_2 \perp \bar{n}_1$$

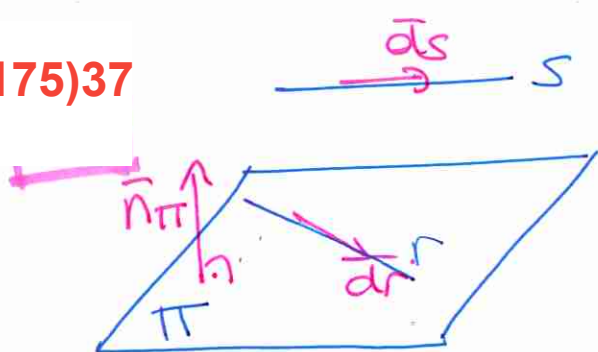
$$P(1, -1, 0) \quad \bar{n}_2 = \bar{dr} \wedge \bar{n}_1 \quad \left. \begin{array}{l} \bar{P} \times (\bar{dr} \wedge \bar{n}_1) = 0 \\ \bar{P} \times \bar{n}_2 = 0 \end{array} \right\}$$

$$\bar{P} \times \begin{vmatrix} \bar{dr} \\ \bar{n}_1 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} x-1 & y+1 & z \\ 1 & 2 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 0 \Rightarrow$$

$$-2(x-1) + 2(y+1) + z - 4z + y + 1 - x - 1 = 0$$

$$\boxed{\pi_2: -3x + 3y - 3z + 6 = 0}$$

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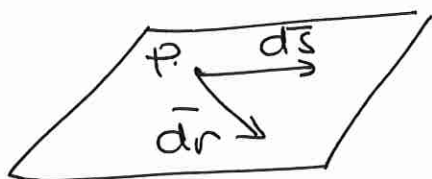


$$S: \begin{cases} x-y-z=0 \\ x-2y+z=0 \end{cases}$$

$$r: \frac{x-1}{2} = y = 2-z \quad P(1,0,2)$$

$$\begin{aligned} \bar{d}s &= (1, -1, -1) \wedge (1, -2, 1) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ 1 & -2 & 1 \end{vmatrix} = \\ &= -\hat{i} - \hat{j} - 2\hat{k} + \hat{k} - \hat{j} - 2\hat{i} = -3\hat{i} - 2\hat{j} - \hat{k} \\ &= (-3, -2, -1) \parallel \underline{\underline{(3, 2, 1)}} \end{aligned}$$

$$\bar{d}r = \underline{\underline{(2, 1, -1)}}$$



Planoaren ek aterotzeko
bi bektore eto $P(1,0,2)$

$$\begin{vmatrix} \bar{P}_x & \bar{d}s & \bar{d}r \\ x-1 & y & z-2 \\ 3 & 2 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 0.$$

$$\begin{aligned} -2(x-1) + 2y + 3(z-2) - 4(z-2) + 3y - (x-1) &= 0 \\ -3x + 3 + 5y - z + 2 &= 0 \end{aligned}$$

$$\boxed{\Pi: -3x + 5y - z + 5 = 0}$$

• Beste modura. $\bar{n}$ eto $P(1,0,2)$.

$$\bar{u} = \bar{d}s \wedge \bar{d}r = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 1 \\ 2 & 1 & -1 \end{vmatrix} = -3\hat{i} + 5\hat{j} - \hat{k}$$

$$\bar{u} \perp \bar{P}_x = 0.$$

$$(-3, 5, -1) \cdot (x-1, y, z-2) = 0. \Rightarrow$$

1. The first part of the paper is a review of the literature on the topic of the effect of the environment on the development of the child. It is a very comprehensive review, covering a wide range of studies and theories. The author discusses the importance of the environment in the development of the child, and how it can influence the child's physical, cognitive, and emotional development. The review is well-organized and easy to read, and it provides a good foundation for the rest of the paper.

2. The second part of the paper is a description of the methods used in the study. The author describes the sample, the data collection methods, and the statistical analysis. The methods are well-described and easy to understand, and they provide a clear picture of how the study was conducted. The author also discusses the strengths and limitations of the methods, and how they might affect the results of the study.

3. The third part of the paper is a presentation of the results of the study. The author presents the data in a clear and concise manner, and it is easy to see the patterns and trends. The results are well-organized and easy to understand, and they provide a clear picture of the effect of the environment on the development of the child. The author also discusses the implications of the results, and how they might be used in practice.

4. The fourth part of the paper is a discussion of the results and their implications. The author discusses the strengths and limitations of the study, and how they might affect the results. The author also discusses the implications of the results, and how they might be used in practice. The discussion is well-organized and easy to understand, and it provides a clear picture of the overall findings of the study.

5. The fifth part of the paper is a conclusion. The author summarizes the main findings of the study, and discusses the implications for future research. The conclusion is well-organized and easy to understand, and it provides a clear picture of the overall findings of the study.

6. The sixth part of the paper is a list of references. The author lists all the studies and theories that were used in the paper, and it is easy to see the range of sources. The references are well-organized and easy to understand, and they provide a clear picture of the background of the study.

